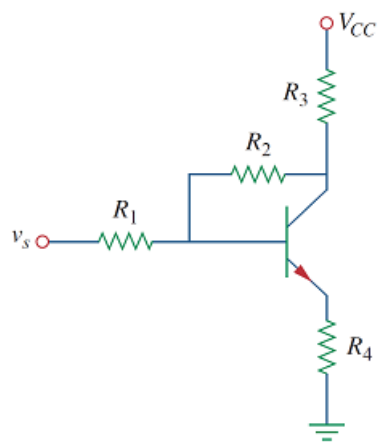


Problem

Using Fig. 3.128, design a problem to help other students better understand transistors. Make sure you use reasonable numbers!

Figure 3.128:



Step-by-step solution

Step 1 of 5

Consider the Figure 1. The value of resistances and the voltages are assumed.

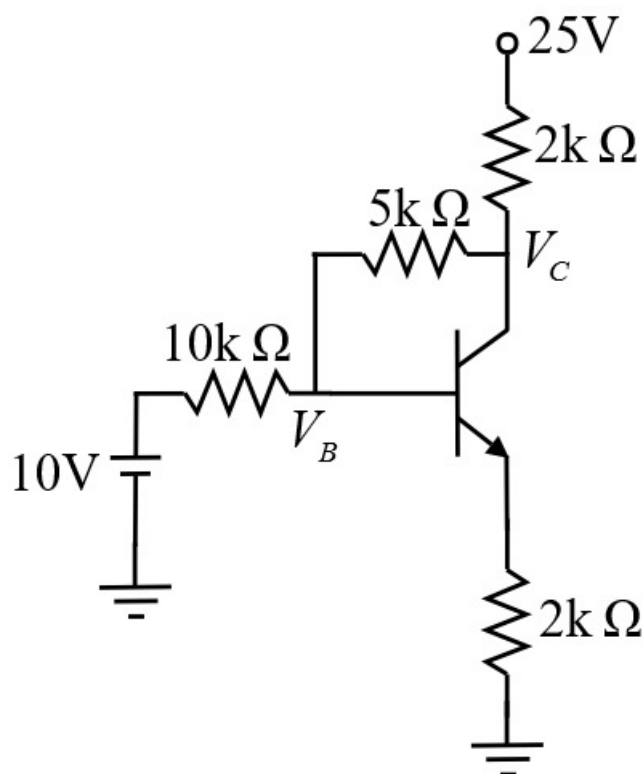


Figure 1

Step 2 of 5

Let the current gain β of the circuit is very high and the voltage V_{BE} is 0.7 V . Then,

$$I_B = 0$$

The current across the emitter I_E is,

$$I_E = I_B + I_C$$

Rewrite the above equation.

$$I_E = I_C \dots\dots (1)$$

Step 3 of 5

Calculate the current I_E .

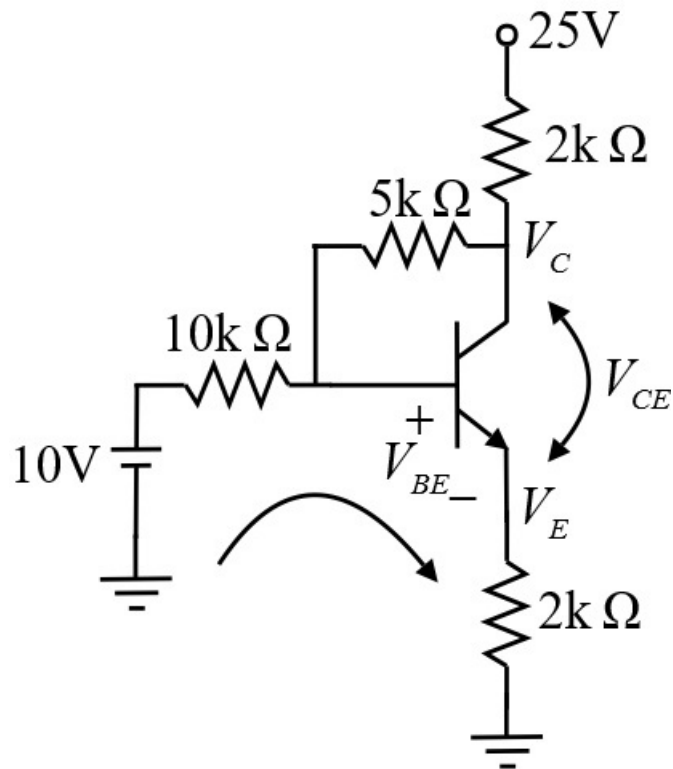


Figure 2

Step 4 of 5

Apply Kirchhoff's voltage law around the loop in Figure 2.

$$-10 + 0.7 + 2 \times 10^3 I_E = 0$$

$$I_E = 4.65 \text{ mA} \dots\dots (2)$$

From, the equations (1) and (2).

$$I_C = 4.65 \text{ mA}$$

Hence, the current I_E and I_C is 4.65 mA.

Calculate the value of V_C .

$$\begin{aligned} V_C &= 25 - (2 \times 4.65) \\ &= 15.7 \text{ V} \end{aligned}$$

Hence, the voltage V_C is 15.7 V.

Calculate the value of V_E .

$$\begin{aligned} V_E &= I_E (2 \times 10^3) \\ &= 4.65 \times 2 \\ &= 9.3 \text{ V} \end{aligned}$$

Hence, the voltage V_E is 9.3 V.

Step 5 of 5

Calculate the value of V_{CE} .

$$\begin{aligned} V_{CE} &= V_C - V_E \\ &= 15.7 - 9.3 \\ &= 6.4 \text{ V} \end{aligned}$$

Hence, the voltage V_{CE} is 6.4 V.